

Exercise Sheet 17

Discrete Mathematics by Hongfei Fu, 2023.12.6

Name: Chen Zhuoyan Student ID: 523030910037 Class: 2302

1. Show that a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.

Answer Area:

- Suppose that a simple graph $G = (V, E)$ is a tree.
 $\therefore$ The graph is a tree.
 $\therefore$ The graph is connected.
For an arbitrary edge $e \in E$, assuming its endpoints are v_1 and v_2 . Since there is only one unique path between v_1 and v_2 , which is e , v_1 and v_2 are not connected after the edge e is removed. That is to say, the graph is not connected.
 $\therefore$ G is connected but the deletion of any of its edges produces a graph that is not connected.
- Suppose that a graph $G = (V, E)$ is connected and the deletion of any of its edges produces a graph that is not connected.
Assuming that there is a simple circuit in the graph and the edge e is in the circuit.
Then after removing the edge e , the graph is still connected since there is a new path between any two vertices by replacing e in the former path (if e is in the former path) with the remaining part of the circuit.
Here is a contradiction. The assumption is wrong.
Therefore there is not a simple circuit in the graph.
 $\therefore$ G is connected.
 $\therefore$ G is a tree.

In conclusion, a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.

2. (2pt) Let G be a simple graph with n vertices. Show that G is a tree if and only if G has no simple circuits and has $n - 1$ edges. [Hint: To show that G is connected if it has no simple circuits and $n - 1$ edges, show that G cannot have more than one connected component.]

Answer Area:

- If $G = V, E$ is a tree.
Then G has no simple circuit and $|E| = |V| - 1 = n - 1$.
- If G has no simple circuits and has $n - 1$ edges.
Assuming that G is not connected and G has s connected components: $G_1 = (V_1, E_1), G_2 = (V_2, E_2), \dots, G_s = (V_s, E_s)$. Every connected component is a tree since it has no simple circuit.
 $\therefore |E| = n - 1 = \sum_{k=1}^s |E_k| = \sum_{k=1}^s (|V_k| - 1) = |V| - s = n - s$.
 $\therefore s \leq 1$. So $s = 1$.
 $\therefore G$ is connected. Hence, G is a tree.

In conclusion, G is a tree if and only if G has no simple circuits and has $n - 1$ edges.

